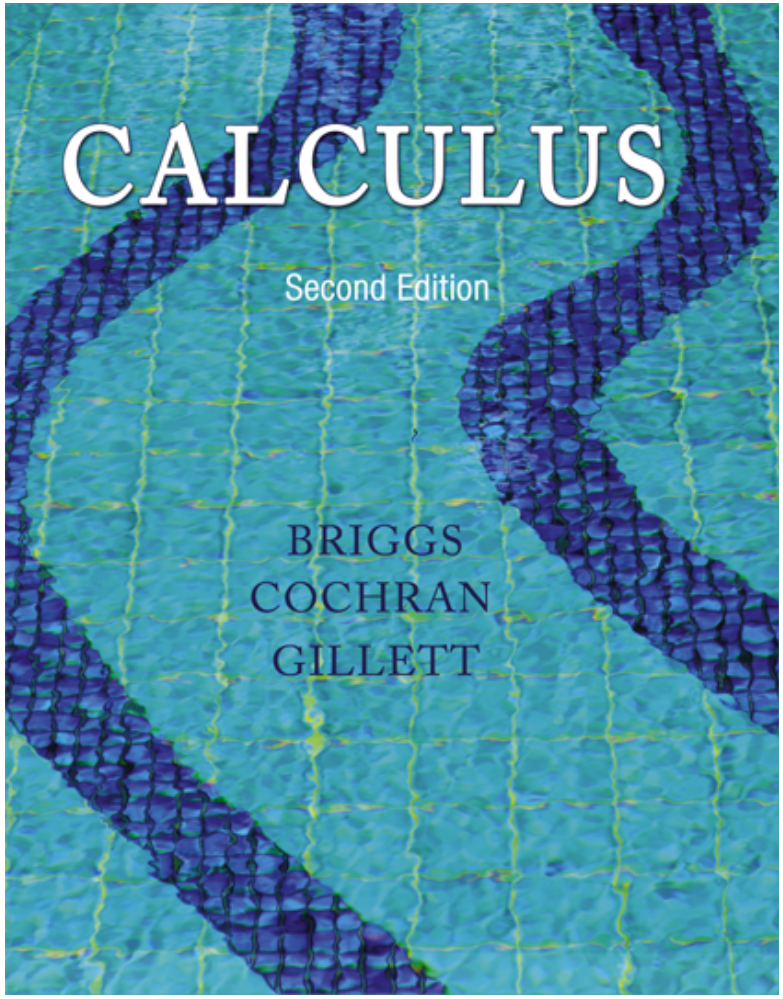
Chapter 9

Sequences and Infinite Series



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9.5

The Ratio, Root, and Comparison Tests

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THEOREM 9.14 Ratio Test

Let $\sum a_k$ be an infinite series with positive terms and let $r = \lim_{k \to \infty} \frac{a_{k+1}}{a_k}$.

1. If $0 \le r < 1$, the series converges.
2. If $r > 1$ (including $r = \infty$), the series diverges.
3. If $r = 1$, the test is inconclusive.

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THEOREM 9.15 Root Test

Let $\sum a_k$ be an infinite series with nonnegative terms and let $\rho = \lim_{k \to \infty} \sqrt[k]{a_k}$.

1. If $0 \le \rho < 1$, the series converges.
2. If $\rho > 1$ (including $\rho = \infty$), the series diverges.
3. If $\rho = 1$, the test is inconclusive.

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THEOREM 9.16 Comparison Test

Let $\sum a_k$ and $\sum b_k$ be series with positive terms.

1. If $0 < a_k \leq b_k$ and $\sum b_k$ converges, then $\sum a_k$ converges.
2. If $0 < b_k \leq a_k$ and $\sum b_k$ diverges, then $\sum a_k$ diverges.

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Figure 9.32

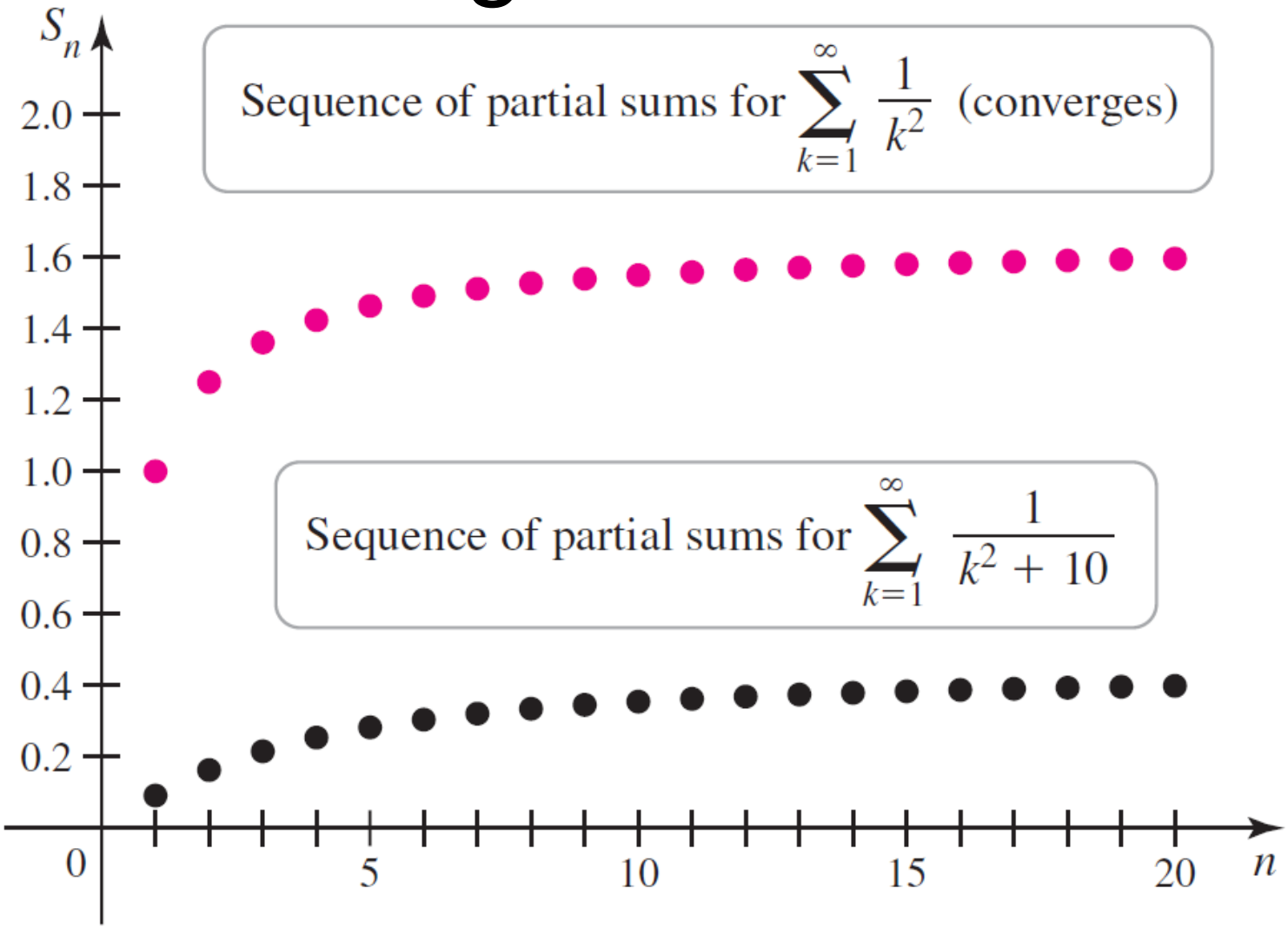

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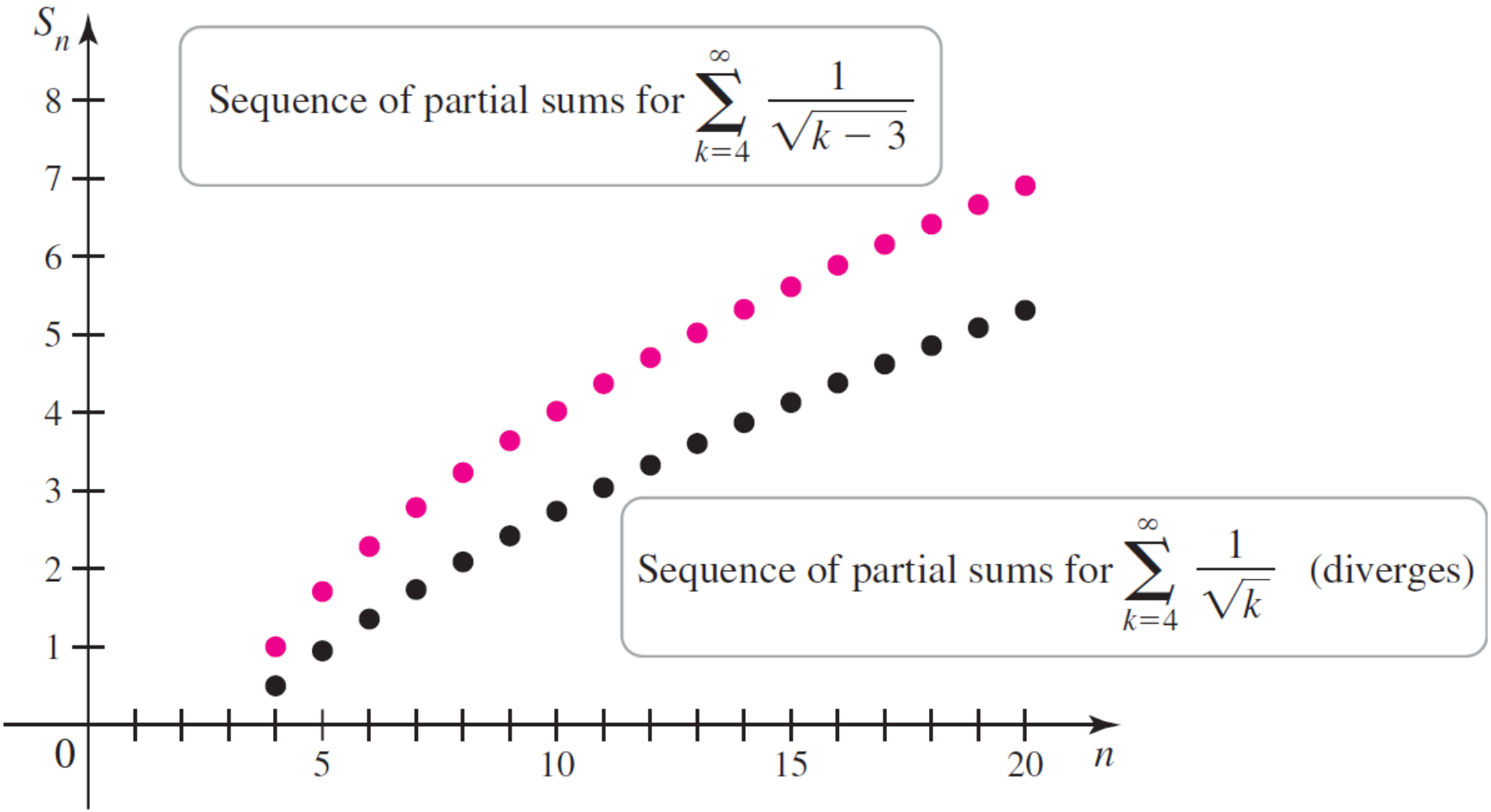
Figure 9.33



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THEOREM 9.17 Limit Comparison Test

Let $\sum a_k$ and $\sum b_k$ be series with positive terms and let

$$\lim_{k \to \infty} \frac{a_k}{b_k} = L.$$

1. If $0 < L < \infty$ (that is, L is a finite positive number), then $\sum a_k$ and $\sum b_k$ either both converge or both diverge.
2. If $L = 0$ and $\sum b_k$ converges, then $\sum a_k$ converges.
3. If $L = \infty$ and $\sum b_k$ diverges, then $\sum a_k$ diverges.

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